ECON 0100 | Fall 2024

MiniExam B | ***Version 3***

This MiniExam will take 15 minutes with quick break to follow. MiniExams are designed to both test your knowledge and challenge you to apply familiar concepts in new environments. Treat it as if you're trying to show me that you understand the material. Answer clearly and completely.

Academic Conduct Code | Name: TAYLOR'S VERSION Student ID: :)

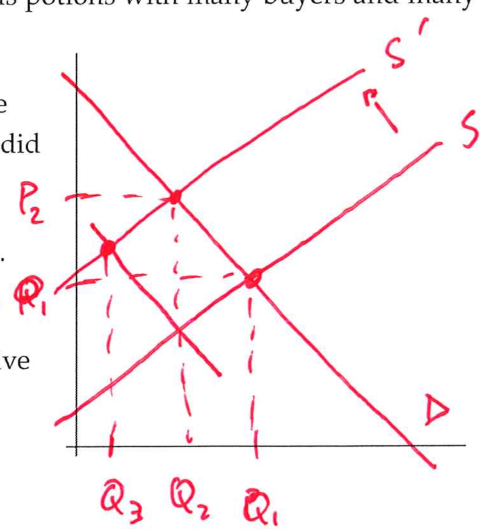
The following academic conduct code is designed to protect the integrity of your work. Print your name/initials beside the five academic honesty agreements. I pledge to my fellow students, the university, and the instructor, that:

TW I will complete this MiniExam solely using my own work.
TW I will not use any digital resources unless explicitly allowed by the instructor.
TW I will not communicate directly or indirectly with others during the MiniExam.

+10 : basically bonus points

Q1. Cauldrons For Days

The Standard Cauldron is a magical tool used widely in the making of various potions with many buyers and many sellers. Use a graph to guide your answers.



a. One terribly cold winter a old mine was destroyed in a cave-in, reducing the availability of the valuable metals used to make the Standard Cauldron. How did the market change after the mining incident?

Which curve shifted? S Price Change: ↑ . Quantity Change: ↓ .

7 — +7: correct, +1: demand
6 — +6: up, correct, +3: down, +1: no change
6 — +6: down, correct, +3: up, incorrect, +1: no change

b. After the mining incident, the Charmfire Toaster became a popular alternative to the Standard Cauldron in many kitchens. How did the market change after both the mining incident and the Charmfire Toaster?

Price Change: Indeterminant . Quantity Change: Down .

6 — +6: correct, +1: incorrect
6 — +6: down, +4: indeterminant, +1: up

Q2. Equilibrim Analysis

For this question, look for answers where students make common mistakes.

After the change in Q1, the demand and supply curves for the Standard Cauldron can be represented by the following relationships. Use a graph to guide your analysis of this market. *No need to find ~~Producer Surplus~~.* Consumer Surplus

$$D : P = 100 - \frac{1}{2}Q_b \qquad (1)$$

$$S : P = 40 + \frac{1}{2}Q_s \qquad (2)$$

a. Equilibrium Price: 70 — 7: correct; 6: small math error; 2: incorrect

b. Equilibrium Quantity: 60 — 6: correct; 5: small...; 2: ...

~~c. Consumer Surplus: 900~~ — 7: ...; 6: ...; 2: ...

c. Producer Surplus: 900

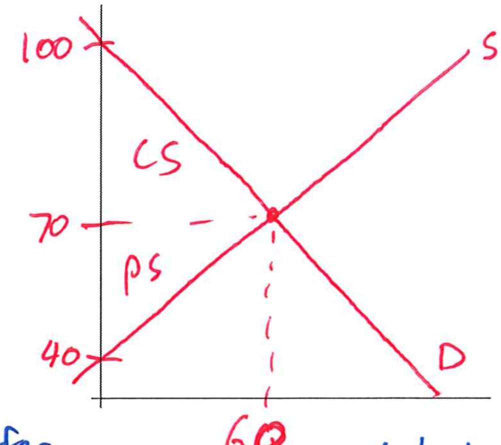
$100 - \frac{1}{2}Q = 40 + \frac{1}{2}Q$

$Q^* = 60$

$P = 40 + \frac{1}{2}\cdot 60$

$P^* = 70$

$CS = 30\cdot 60\cdot \frac{1}{2} = 900$



Q3. Cauldron Conspiracy

For this question, look for answers where students make common mistakes.

In a much debated move, the Ministry of Magic imposed a price ceiling of 50 galleons on the Standard Caulron market in a misguided attempt at ensuring fair prices for everyday households. Use a graph to guide your discussion of the impact of this price control on the market. *No need to find ~~Producer Surplus~~.* Consumer Surplus

a. Price: 50 — 7 — Same grading system

b. Quantity Supplied: 20 — 6

c. Quantity Demanded: 100 — 6

b. Quantity Exchanged: 20 — 6

e. ~~Consumer Surplus~~ Producer Surplus: 100 — 7

f. Surplus or Shortage (*circle one*)

7: 7: correct; 2: incorrect

$50 = 100 - \frac{1}{2}Q_b$

$Q_b = 100$

$50 = 40 + \frac{1}{2}Q_s$

$20 = Q_s$

$PS = 10\cdot 20\cdot \frac{1}{2} = 100$